Explicit finite extraction from convolution profiles for matrix multiplication

Abstract

The convolution-profile argument for the matrix multiplication exponent $9/4$ passes from finite tensor relations to an existence statement for fast algorithms. We give a direct finite extraction from those relations. An explicit positive rational combination of neighboring-length and three-sector relations telescopes to a diagonal profile bound. Clearing denominators gives integer tensor types. We compose their exact restrictions with a Fourier separator, exact interpolation, repeated use of a fixed catalyst, and a batched recursion. All direct-sum multiplicities and finite overheads are retained. For a profile index a, the resulting exponents approach $3\log(2a-1)/\log F(a,a)$, where $F(a,a) = \frac{3a-1}{2}\prod_{h=1}^{a-1}(1+1/(3h))$. These bounds tend to $9/4$. An integer-arithmetic selection procedure therefore produces, for each positive rational ε, a finite algorithm with arithmetic complexity $O_\varepsilon(n^{9/4+\varepsilon})$. It also specifies a rational bilinear seed with exponent below $9/4+\varepsilon$. We prove that F is the least profile satisfying the underlying scalar axioms and determine the exact denominator scale of the rational certificate. That scale grows at least exponentially in a. The construction resolves the finite extraction problem for these relations, but supplies neither an exact exponent-$9/4$ recursion nor a practical algorithm.

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1 Introduction

A bound on the matrix multiplication exponent and a finite description of an algorithm are closely related, but they are different outputs of a proof. An asymptotic tensor argument may establish the existence of a suitable rank decomposition without giving its dimension or coefficients. A finite extraction identifies the tensor maps, chooses their parameters, and accounts for every cost needed to turn them into an arithmetic algorithm. This paper carries out that extraction for the convolution relations used in the preprint *An Upper Bound of* $9/4$ *for the Matrix Multiplication Exponent* [10].

That preprint introduces a separation of tensor blocks that share one leg, applies it to polynomial multiplication, and derives two inequalities for a symmetrized convolution profile. The profile grows at least as $a^{4/3}$ on the diagonal. A detecting-character argument then gives the claimed bound $\omega \le 9/4$, where ω is the infimum of arithmetic exponents for square matrix multiplication. The last algorithmic step chooses a matrix size and an exact rank decomposition whose existence follows from that bound. It does not specify the chosen decomposition.

We replace this last passage by a direct construction. The starting point is an exact solution of the scalar profile problem. Its affine rows determine rational probabilities for the two tensor relations. Explicit positive rational multipliers make these relations telescope, leaving a diagonal term and a boundary term in the formal resource identity. Clearing denominators turns this identity into integer tensor types. We then construct the associated maps and derive a recursive algorithm with all finite costs included.

1.1 Main results

The principal conclusion is the following effective form of the exponent bound. A rational bilinear seed means a finite list of rational linear forms whose products and recombinations compute a square matrix product.

Theorem 1.1. *For every positive rational* ε*, a terminating procedure produces an integer* $d \ge 2$ *and an exact rational bilinear algorithm for* $d \times d$ *matrix multiplication using fewer than* $d^{9/4+\varepsilon}$ *bilinear multiplications, each multiplying a linear form in the left input by a linear form in the right input. A fixed recursion on this seed multiplies* $n \times n$ *matrices over any field of characteristic zero in* $O_\varepsilon(n^{9/4+\varepsilon})$ *arithmetic operations.*

In particular, $\omega \le 9/4$. A bound with any positive real allowance follows by choosing a smaller positive rational allowance.

The procedure first gives a relative algorithm: a fixed number of larger matrix products are reduced simultaneously to a fixed number of smaller ones. A padded batch recursion makes this

a usual arithmetic algorithm. Unrolling that recursion and restricting coefficients from its fixed cyclotomic field to $\mathbb{Q}$ gives the seed in Theorem 1.1. The parameter choices use exact integer comparisons, so the statement does not depend on a numerical tolerance or an unspecified rank search. The complete proof is in Theorem 5.5 and Corollary 5.7. For allowances at most $3/4$, the prescribed seed generator returns a dimension larger than the relative scheme's recursive input dimension.

Three intermediate results describe both the construction and its limits. First, the least positive symmetric profile satisfying normalization, discrete concavity, and shifted tripling is

$$F(m,n) = s_{\min(m,n)} \left(\max(m,n) + \frac{\min(m,n)-1}{2} \right), \qquad s_a = \prod_{h=1}^{a-1} \left(1 + \frac{1}{3h} \right).$$

This gives the exact scalar bound at every pair of indices, rather than only its diagonal growth rate. Second, we give closed formulas for the positive rational multipliers and prove the resulting integer resource identity. Third, we determine their minimal common denominator and show that it grows at least exponentially with the profile index. The diagonal certificate uses only tripling at the row index; the full least-profile theorem allows all shifted tripling inequalities.

For fixed index a, the constructed exponents approach

$$\tau_a = \frac{3\log(2a-1)}{\log F(a,a)}.$$

Every τ_a is strictly larger than $9/4$, and these values tend to $9/4$ as a increases. At each finite stage the cost formula retains the source copies, output multiplicity, interpolation, catalyst supply, and rounding of the recursive branch count. We also reduce the shared-leg separator from the preprint's $5M$ source copies to $2M-1$, and give a separate exact two-power projection.

The numerical exponent $9/4$ is already the conclusion announced in OpenAI [10]. Our contribution is the explicit finite extraction and its accompanying sharp scalar and finite-cost analysis. The proof below is self-contained at the level of tensor maps. It uses neither the detecting-character theorem nor the spectral appendix of that preprint.

1.2 Relation to previous work

Strassen's recursive use of a finite bilinear identity established the basic link between small matrix algorithms and an arithmetic exponent below three [14]. Approximate bilinear algorithms and their conversion to exact algorithms by tensor powers and interpolation form another part of the foundation [3]. Schönhage's asymptotic sum inequality relates the simultaneous computation of independent matrix products to the exponent [12]. An accessible detailed account of finite interpolation and the sum inequality is given by Stothers [13, Theorem 7 and its preceding discussion]. Our root-of-unity averaging is an explicit instance of this interpolation principle. Our batch recursion keeps the simultaneous-output count visible instead of invoking the sum inequality as a black box.

The asymptotic spectrum provides a general language for comparing tensor powers through numerical invariants [5, 16]. Relative bilinear complexity is likewise an established setting for reductions between matrix multiplication tensors [15]. Catalytic tensor comparisons also occur in the study of matrix multiplication: an auxiliary tensor is retained on both sides of a reduction [4]. The present construction starts from specific finite relations within this framework. Its rational certificate supplies the multiplicities needed to compose those relations, so the proof of the algorithm does not require an abstract separation argument.

Fixed-frequency restrictions and entropy estimates are standard in tensor-power methods. Le Gall's analysis of tensor powers gives a detailed example of these methods and of balancing products

over leg permutations [9]. Recent upper bounds based on refinements of the laser method also rely on carefully chosen distributions and asymmetric tensor powers [1, 2, 8]. For numerical context, the SODA 2025 paper of Alman et al. [2] proves $\omega < 2.371339$, while the 2026 preprint of Dupont et al. [8] reports $\omega < 2.371177$. The 9/4 bound announced in the October 2026 preprint [10] is substantially stronger; our proof establishes the finite tensor route to that bound independently of the preprint's spectral conclusion. The construction uses a specific family of convolution relations and a closed rational certificate for it. Interpolation, entropy counting, symmetrization, and recursive use of bilinear identities are established tools.

The closest source for the actual tensor maps is OpenAI [10], Proposition 3.1 and Section 4. We give the bases and coordinate maps explicitly, strengthen its two initial degenerations to exact type restrictions, and improve its separator count. The least-profile formula, the rational telescoping weights, their denominator formula, and the complete finite cost expression are the specific developments established here. These are comparisons with an identified prior argument; no claim of a first general constructive conversion of asymptotic tensor bounds is needed.

In particular, an existence theorem for a suitable complex rank decomposition already implies an eventual symbolic search: its defining polynomial equations have rational coefficients, and a complex solution can be replaced by an algebraic one. The present procedure has a different purpose. It generates identities by prescribed maps, with formulas for their finite resource counts, rather than enumerating candidate solutions to rank equations. Its value is mathematical control of that passage, not a practical search bound.

1.3 Scope and organization

The construction concerns exact arithmetic. It has very large fixed constants, no established numerical stability bound, and no demonstrated useful matrix-size range. The cost analysis gives a reason for this size: the exact rational types alone have exponentially growing normalization. For the prescribed index-two batch schedule, even parameters that beat $\log_4 48$ require a recursive input dimension with more than 11.37 billion decimal digits. This is a limitation of the specified schedule, not of arbitrary algorithms with the same exponent.

Section 2 fixes the tensor conventions and the arithmetic model. Section 3 proves the least-profile theorem and the rational certificate. Section 4 gives the type maps and separator. Section 5 composes them, proves the batched recursion, and specifies terminating parameter choices. Section 6 analyzes the denominator growth and finite example. Section 7 describes the consequences and the limits of the method. The appendices specify the full composed relation maps, prove the optional separator refinements, and give the detailed lower bound for the small-index schedule.

2 Tensors, restrictions, and arithmetic algorithms

All logarithms are natural. All vector spaces are finite dimensional. We first work over $\mathbb{C}$; every coefficient used in the construction will be algebraic. A tensor is written as a trilinear form in three disjoint variable groups, called its legs. Its rank $\mathrm{R}(T)$ is the least number of products of three linear forms whose sum is T. We write $A \succeq B$ if substitutions that are linear on each leg transform A into B. Such a restriction gives a bilinear algorithm for B using a bilinear algorithm for A. If the substitutions depend on a nonzero parameter t, and the transformed tensor is a polynomial $B + tB_1 + \cdots + t^D B_D$, we write $A \trianglerighteq B$. The individual substitutions may be Laurent polynomials; only their combined tensor is required to have no negative powers. Every degeneration below is supplied with its substitutions and a degree bound. No general approximation theorem is needed to use it.

For a positive integer s, the notation sT means the full direct sum of s copies of T, with independent variables on all three legs. It does not mean scalar multiplication of the coefficients. Tensor products distribute over these direct sums. We abbreviate tensor products by juxtaposition when no confusion can arise. We never cancel a direct-sum multiplicity or a tensor factor from a restriction.

Lemma 2.1 (Composition rules used below)**.** *Suppose a triple of Laurent-polynomial linear substitutions $\Phi_i(t)$ transforms a tensor A_i into $G_i(t) = \sum_{d=0}^{D_i} t^d G_{i,d}$. Let P_i be a specified restriction with $P_i(C_i) = A_i$, and let R_i be a triple of linear maps on the whole target variable spaces. If P_i, R_i are independent of t, then*

$$R_i \Phi_i(t) P_i(C_i) = \sum_{d=0}^{D_i} t^d R_i(G_{i,d}).$$

Their tensor product has degree at most $\sum_i D_i$. Its constant coefficient is $\bigotimes_i R_i(G_{i,0})$. Permuting the three legs preserves these statements.

Proof. The first identity follows by substitution and linearity on the tensor space. In a tensor product of polynomials with nonnegative exponents, degrees add, and a term has degree zero exactly when every factor does. Permuting the legs only permutes the three substitution maps. □

The square matrix multiplication tensor is

$$\mathsf{M}_d = \sum_{i,j,k=0}^{d-1} x_{ij} y_{jk} z_{ki}.$$

Its third variables encode the output coordinates by duality. Let $B_X(d) = x \sum_{i=0}^{d-1} y_i z_i$, and define $B_Y(d)$ and $B_Z(d)$ by permuting the legs. Relabeling coordinates gives

$$\mathsf{M}_d \mathsf{M}_e \cong \mathsf{M}_{de}, \qquad B_X(d) B_Y(d) B_Z(d) \cong \mathsf{M}_d. \tag{1}$$

For example, the three paired indices in the second product can be taken as the row, inner, and column indices of matrix multiplication. Consequently the product of all six leg permutations of $B_X(d)$ is M_{d^2}.

The basic convolution tensor is

$$C(r,b) = \sum_{i=0}^{r-1} \sum_{j=0}^{b-1} x_i y_j z_{i+j}. \tag{2}$$

It represents multiplication of polynomials of degrees at most $r-1$ and $b-1$. Put $d = r+b-1$ and let ξ be a primitive d-th root of unity. Evaluation at ξ^ℓ gives the explicit rank decomposition

$$C(r,b) = \frac{1}{d} \sum_{\ell=0}^{d-1} \left(\sum_i \xi^{\ell i} x_i \right) \left(\sum_j \xi^{\ell j} y_j \right) \left(\sum_{h=0}^{d-1} \xi^{-\ell h} z_h \right). \tag{3}$$

Indeed $0 \le i+j \le d-1$, so the Fourier sum selects exactly $h = i+j$. The output flattening has rank d, and hence $\mathrm{R}(C(r,b)) = d$.

For $\pi \in S_3$, superscript π denotes a permutation of the legs. Define the balanced tensor

$$Q_{r,b} = \bigotimes_{\pi \in S_3} C(r,b)^\pi. \tag{4}$$

The six factors are kept as labeled factors, even when some are isomorphic. Swapping the inputs in (2) and reordering these factors gives

$$Q_{r,b} \cong Q_{b,r}, \qquad \mathrm{R}(Q_{r,b}) \le (r+b-1)^6, \qquad Q_{1,2} \cong \mathsf{M}_4. \tag{5}$$

All these isomorphisms are coordinate permutations.

Arithmetic complexity counts scalar additions, subtractions, and multiplications, including multiplication by fixed coefficients. The algorithm may depend on a fixed exponent allowance ε, but not on the input dimension n. Its constant preprocessing cost may depend on ε. This is an arithmetic-operation statement, not a bit-complexity or floating-point stability statement.

Lemma 2.2 (Exactification)**.** *Suppose specified substitutions transform A into a polynomial $G(t) = \sum_{h=0}^{D} t^h G_h$. For every positive integer k,*

$$(kD+1)A^{\otimes k} \succeq G_0^{\otimes k}.$$

Proof. Set $E = kD+1$ and take a primitive E-th root η. The tensor polynomial $G(t)^{\otimes k}$ has degree at most kD. Thus

$$\frac{1}{E}\sum_{h=0}^{E-1} G(\eta^h)^{\otimes k} = G_0^{\otimes k}.$$

Each evaluation uses one copy of $A^{\otimes k}$; identifying its output variable groups and absorbing $1/E$ into one leg gives the claimed restriction. Every evaluation point is nonzero, so Laurent substitutions cause no problem. □

3 The least scalar profile and an explicit rational certificate

We first isolate the scalar problem. The connection to actual tensors will be proved by direct maps in Sections 4–5. In particular, no assumption that the least scalar profile is a tensor character enters the construction.

Definition 3.1. A profile is a positive symmetric function $P : \mathbb{N}_{>0}^2 \to \mathbb{R}$ satisfying

$$P(1,b) = b, \tag{6}$$

$$2P(a,b) \ge P(a,b-1) + P(a,b+1) \quad (b \ge 2), \tag{7}$$

$$P(a,3h+a-1) \ge 3P(a,h) \quad (h \ge 1). \tag{8}$$

Define

$$s_m = \prod_{h=1}^{m-1}\left(1+\frac{1}{3h}\right), \qquad F(m,n) = s_{\min(m,n)}\left(\max(m,n) + \frac{\min(m,n)-1}{2}\right). \tag{9}$$

An empty product equals one.

Theorem 3.2 (Least profile)**.** *The function F is a profile, and every profile P satisfies $P(m,n) \ge F(m,n)$. Moreover,*

$$F(m,n) \le (m+n-1)^{4/3}, \qquad \log F(a,a) = \frac{4}{3}\log a + O(1). \tag{10}$$

The first inequality is strict when $m = n > 1$.

Proof. Normalization and symmetry are immediate. Using $s_{b+1}/s_b = (3b+1)/(3b)$ and $s_{b-1}/s_b = (3b-3)/(3b-2)$, direct subtraction gives

$$2F(a,b) - F(a,b-1) - F(a,b+1) = \begin{cases} \dfrac{2(a-b)s_b}{3b(3b-2)}, & 2 \le b \le a, \\ 0, & b > a. \end{cases}$$

Thus each row is concave. Its affine tail has slope s_a, so its extension bounds the row from above: $F(a,h) \le s_a(h + (a-1)/2)$. Evaluating this affine tail at $3h + a - 1 \ge a$ proves (8) for F.

For minimality, write $D_a = P(a,a)$, $k_a = 2D_a/(3a-1)$, and $\Delta_{a,h} = P(a,h+1) - P(a,h)$. Tripling at $h = a$ and decreasing increments imply

$$2D_a \le P(a,4a-1) - P(a,a) \le (3a-1)\Delta_{a,a}.$$

By symmetry and concavity, for $a \ge 2$,

$$D_a - D_{a-1} = \Delta_{a-1,a-1} + \Delta_{a,a-1} \ge k_{a-1} + k_a.$$

It follows that $k_a \ge (3a-2)k_{a-1}/(3a-3)$. Since $k_1 = 1$, this proves $k_a \ge s_a$ and the diagonal bound.

Fix $m < n$. Starting with $x_0 = m$, put $x_{j+1} = 3x_j + m - 1$, and choose j with $x_j \ge n$. Then

$$x_j - m = (3^j - 1)(3m-1)/2, \qquad P(m, x_j) \ge 3^j D_m.$$

A concave sequence lies above the chord joining two of its entries. The chord from m to x_j therefore gives

$$P(m,n) \ge D_m + \frac{n-m}{x_j - m}(3^j - 1)D_m = \frac{2D_m}{3m-1}\left(n + \frac{m-1}{2}\right) \ge F(m,n).$$

Only finitely many profile inequalities have been used.

Induction gives $s_a^3 \le (3a-1)/2$, strictly for $a > 1$. The induction step reduces to $(3a+2)(3a)^3 - (3a-1)(3a+1)^3 = 6a + 1 > 0$. For $m \le n$, both $(3m-1)/2$ and $n + (m-1)/2$ are at most $m + n - 1$. Cubing (9) proves (10) and its strict diagonal version. Finally, $\log(1+x) = x + O(x^2)$ and convergence of $\sum h^{-2}$ give $\log s_a = \frac{1}{3}\log a + O(1)$. □

The theorem makes precise a limitation of the scalar argument: these axioms, even together with $P(a,b) \le (a+b-1)^{4/3}$, cannot force diagonal growth faster than $a^{4/3}$. This is a statement about the scalar relaxation, not a claim that F comes from a tensor character. The off-diagonal minimality uses tripling at all h. The diagonal bound and the rational certificate below need only the instances with h equal to the row index, which are the instances realized by our tensor maps.

3.1 A certificate with fixed rational probabilities

For a fixed target index $a \ge 2$, use rows $2 \le r \le a$ and columns $r \le b \le 4r-2$. The affine formula in (9) holds in row r throughout $b \ge r-1$: at its left endpoint this follows from $s_r/s_{r-1} = (3r-2)/(3r-3)$. Set

$$q^{\pm}_{r,b} = \frac{2b + r - 1 \pm 2}{2(2b + r - 1)}. \tag{11}$$

These positive rational probabilities equal the neighboring values of F divided by $2F(r,b)$. Put $y_{r,b} = \log(P(r,b)/F(r,b))$. Weighted arithmetic–geometric mean applied to (7) yields

$$y_{r,b} - q^{-}_{r,b}y_{r,b-1} - q^{+}_{r,b}y_{r,b+1} \ge 0. \tag{12}$$

Indeed the normalized concavity inequality is

$$\frac{P(r,b)}{F(r,b)} \geq q^-_{r,b}\frac{P(r,b-1)}{F(r,b-1)} + q^+_{r,b}\frac{P(r,b+1)}{F(r,b+1)} \geq \exp\big(q^-_{r,b}y_{r,b-1} + q^+_{r,b}y_{r,b+1}\big).$$

We will also use the equality $H(q)+\sum q^{\pm}\log F(r,b\pm1) = \log(2F(r,b))$, where $H(q) = -\sum q^{\pm}\log q^{\pm}$. Tripling at $h = r$ gives $y_{r,4r-1} - y_{r,r} \geq 0$.

Put $L = 4r-1$ and $D_r = 3r-1$. In the final row $r = a$, set

$$\lambda_{r,b} = \frac{2(2b+r-1)(L-b)}{3(r-1)D_r}, \qquad \mu_r = \frac{1}{r-1}. \tag{13}$$

In each lower row $r < a$, set

$$\begin{aligned} \lambda_{r,r} &= \frac{2D_r}{3(r-1)}, \\ \lambda_{r,b} &= \frac{4(2b+r-1)(L-b)}{3(r-1)(3r+1)} \quad (b > r), \\ \mu_r &= \frac{2D_r}{(r-1)(3r+1)}. \end{aligned} \tag{14}$$

Proposition 3.3 (Rational telescoping)**.** *Weight (12) by $\lambda_{r,b}$ and tripling by μ_r. Their sum over all specified rows and columns is exactly $y_{a,a} - y_{1,2} \geq 0$. All weights and all split weights $\lambda q^{\pm}$, $\mu/3$ are positive rationals.*

Proof. Write $h_b = 2b+r-1$ and $f_b = \lambda_{r,b}/h_b$. At an interior column the concavity contribution is $h_b(f_b - (f_{b-1}+f_{b+1})/2)$. In the final row $f_b = 2(L-b)/(3(r-1)D_r)$ is affine. The left external coefficient is $-h_{r-1}f_r/2 = -1$. At r, the concavity contribution is $r/(r-1)$; subtracting $\mu_r = 1/(r-1)$ leaves 1. The terminal coefficient is $-h_L f_{L-1}/2 + \mu_r = 0$. Thus this row leaves $y_{r,r} - y_{r,r-1}$.

In a lower row, f_b is affine for $b > r$ and $f_r = 2/(3(r-1))$. Its affine extension from $b > r$ differs at r by $2/(3r+1)$. The second difference therefore leaves coefficient 1 at $r+1$. The left external coefficient remains -1; the concavity coefficient at r is $2(3r-1)/((r-1)(3r+1)) = \mu_r$, which cancels with $-\mu_r$, and the terminal coefficient cancels with μ_r. This row leaves $y_{r,r+1} - y_{r,r-1}$. Symmetry identifies $y_{r,r+1} = y_{r+1,r}$, so all row boundaries cancel except $y_{a,a} - y_{2,1} = y_{a,a} - y_{1,2}$. □

The probabilities in this certificate are fixed by F, not by the unknown values of P. This fact is what permits an exact finite tensor construction.

3.2 Integer types and their entropy identity

Let $K = K_a$ be the least common multiple of the denominators of all λq^+, λq^-, and $\mu/3$, in lowest terms. For each concavity relation write

$$w = K\lambda, \qquad u = K\lambda q^+, \qquad d = K\lambda q^-; \qquad w = u + d.$$

For each tripling relation write $z = K\mu$; then $3 \mid z$. Let

$$U = \sum w, \quad T = \sum z, \quad I_c = \frac{3a^2+a-4}{2}, \quad I_t = a-1, \quad I = \frac{3(a+2)(a-1)}{2}. \tag{15}$$

The sums run over concavity and tripling relations, respectively. There are I_c and I_t such relations.

For a formal symbol $[r,b]=[b,r]$, Proposition 3.3 gives the exact integer identity

$$\sum_{r,b}\bigl(w[r,b]-u[r,b+1]-d[r,b-1]\bigr)+\sum_{r} z\bigl([r,4r-1]-[r,r]\bigr)=K[a,a]-K[1,2]. \tag{16}$$

This is an identity of exponent vectors, not an identity of sums of tensors. Let $i_{r,b}$ and $o_{r,b}$ be the nonnegative source and output counts on its left, after identifying symmetric pairs. Put $c_{r,b}=\min(i_{r,b},o_{r,b})$ and

$$\mathcal{C}=\bigotimes_{r\le b} Q_{r,b}^{\otimes c_{r,b}}, \qquad \mathcal{C}_N=\mathcal{C}^{\otimes N}. \tag{17}$$

Thus $\sum c_{r,b}=U+T-K$. The common factors are retained as the catalyst $\mathcal{C}$; they are not formally canceled from a restriction.

Tightness of each entropy relation at F also gives

$$2^{K-U}3^{T}\prod_{\text{concavity relations}}\frac{w^w}{u^u d^d}=F(a,a)^K. \tag{18}$$

To check the identity, multiply $(2F(r,b))^w=(w^w/(u^u d^d))F(r,b+1)^u F(r,b-1)^d$ and $F(r,4r-1)^z=3^z F(r,r)^z$ over the relations. Use (16) and $F(1,2)=2$. This derivation proves the identity over the rationals, without a numerical entropy calculation.

4 Exact type restrictions and separation of a shared leg

We now realize the two families of relations by finite tensor maps. There are two distinct tasks. A type projection removes off-diagonal blocks, but leaves the first leg shared. A subsequent separator makes the surviving blocks independent on all three legs. Precisely, matched sectors sharing the first leg have the form $\sum_h A_h$, with $A_h\in X\otimes Y_h\otimes Z_h$ and direct-sum decompositions on the second and third legs. All supported terms use the same sector label on those two legs, while the space X is common to all sectors.

4.1 The neighboring-length map

Let V_e be the homogeneous binary forms of degree e in u,v, and let $W=\operatorname{span}(s,w)$. For degree $e\ge 1$, put $s_i=u^{e-i}v^i s$ and $w_i=u^{e-i}v^i w$. Choose quotient lifts $s_0,\dots,s_e,w_e$ and kernel vectors $k_i=w_i-s_{i+1}$, $0\le i<e$. These form a basis of $V_e\otimes W$: substitution $(s,w)=(u,v)$ sends the quotient lifts to distinct monomials and kills the kernel vectors, which are independent. The kernel is $(uw-vs)V_{e-1}$.

Lemma 4.1 (Exact neighboring-length types)**.** *For positive integers n_+,n_- with $n=n_++n_-$, a specified restriction of $(C(r,b)B_X(2))^{\otimes n}$, $r,b\ge 2$, has $\binom{n}{n_+}$ matched sectors on its second and third legs. Each sector is isomorphic to $C(r,b+1)^{\otimes n_+}C(r,b-1)^{\otimes n_-}$. The first leg is still shared; the sector-specific coordinate permutation is applied only after separation.*

Proof. View $C(r,b)B_X(2)$ as multiplication $V_{r-1}\times(V_{b-1}\otimes W)\to V_{r+b-2}\otimes W$. Use the preceding bases on its second-input and output spaces, and the dual output basis on its third leg. Multiplication by $u^{r-1-i}v^i$ shifts each monomial index by i. It preserves the kernel and induces ordinary multiplication on the quotient. The two diagonal blocks are $C(r,b+1)$ and $C(r,b-1)$. The only off-diagonal terms are

$$x_i q_b k^*_{b-1+i}, \qquad 0\le i<r-1, \tag{19}$$

where $q_b = w_{b-1}$ is the last input quotient lift and the star denotes an output-dual coordinate. Indeed, the image of that lift has index $b-1+i$. Unless this is the last output index, $w_{b-1+i} = s_{b+i} + k_{b-1+i}$. All other quotient lifts stay in the quotient, and a kernel vector maps to a kernel vector. This proves (19).

In n tensor positions, retain words with exactly n_- kernel labels on both the second and third legs. Each off-diagonal term increases the output kernel count by one, and no term decreases it. Equal retained counts therefore remove every off-diagonal term. In a surviving term the two sector labels agree at each position. Choosing the n_+ quotient positions gives the asserted number of sectors and their tensors. □

This is an exact restriction after a fixed basis change. In particular, it incurs no interpolation degree.

4.2 The three-sector map

Lemma 4.2 (Exact three-sector types)**.** *For $r \geq 2$ and $3 \mid n$, a coordinate restriction of $C(r, 4r-1)^{\otimes n}$ has $n!/(n/3)!^3$ matched sectors on its second and third legs. Each sector contains $2n/3$ factors isomorphic to $C(r, r)$ and $n/3$ factors isomorphic to $C(r, r)$ with its second and third legs exchanged. The first leg remains shared. The sector-specific coordinate permutations are applied only after separation.*

Proof. A supported term has $0 \leq i < r$, $0 \leq y < 4r-1$, and $z = i + y$. Partition the index sets into the following intervals, with endpoints included:

	left	middle	right
Y	$[0, r-1]$	$[r, 3r-2]$	$[3r-1, 4r-2]$
Z	$[0, 2r-2]$	$[2r-1, 3r-2]$	$[3r-1, 5r-3]$.

A left Y-index forces a left Z-index, and a right Y-index forces a right Z-index. Every unmatched term therefore has a middle Y-label and an outer Z-label. Retain $n/3$ labels of each kind on both legs in a tensor power. Equality of the middle counts eliminates every unmatched term; the remaining labels agree positionwise.

The outer blocks are translates of $C(r, r)$. The middle block has terms

$$x_{r-1-i}\, y_{r+i+j}\, z_{2r-1+j}, \qquad 0 \leq i, j < r.$$

Reversing the first index and exchanging the last two legs identifies it with $C(r, r)$. These identifications are statements about each sector separately. We do not apply different first-leg maps to blocks while that leg is shared. □

4.3 A Fourier separator

The following form includes an auxiliary dimension and a phase slope. The main construction uses only $K_0 = M$, $c = 1$.

Proposition 4.3 (Shared-leg separator)**.** *Let $A = \sum_{h=0}^{M-1} A_h$, with $A_h \in X \otimes Y_h \otimes Z_h$, where the Y_h and Z_h are pairwise disjoint sectors. For positive integers K_0, c, put $L = K_0 + c(M-1)$. There is an explicit degeneration*

$$LA \trianglerighteq \bigoplus_{h=0}^{M-1} (A_h B_X(K_0)). \tag{20}$$

Its polynomial has exponent support contained in $\{cd^2 : 0 \le d \le d_*\}$*, where* $d_* = \min(M-1, \lfloor (K_0-1)/c \rfloor)$*. In particular,* $K_0 = M, c = 1$ *uses* $2M-1$ *source copies and has degree at most* $(M-1)^2$*. Permuting all three legs gives the same statement with any chosen shared leg.*

Proof. Choose a primitive L-th root ζ. In source copy ℓ, $0 \le \ell < L$, make the substitutions

$$x_a^{(\ell)} \longmapsto \sum_{g=0}^{M-1} \zeta^{\ell c g} X_{a,g},$$
$$y_{h,b}^{(\ell)} \longmapsto \sum_{u=0}^{K_0-1} \zeta^{\ell(u-ch)} Y_{h,b,u},$$
$$z_{h,d}^{(\ell)} \longmapsto L^{-1} \sum_{v=0}^{K_0-1} \zeta^{-\ell v} Z_{h,d,v}.$$

Summing the copies retains exactly $u - v + c(g-h) \equiv 0 \pmod{L}$. Its absolute value is at most $L-1$, so the congruence is the equality $u - v + c(g-h) = 0$. Multiply the new variables by the respective powers of t

$$t^{cg^2}, \qquad t^{2hu-ch^2}, \qquad t^{-2hv}.$$

Every surviving term has total exponent $cg^2 + 2h(u-v) - ch^2 = c(g-h)^2$. It is zero exactly when $g = h$ and $u = v$. These terms are precisely the target in (20), whose first variables $X_{a,h}$ are now independent for different h. Moreover $c|g-h| = |u-v| \le K_0 - 1$, proving the exponent support. □

Only after this separator has been applied do we reorder tensor positions or use sector-specific first-leg reversals. Applying the maps on all six permutations balances the exchanged factors in Lemma 4.2: composition with the transposition σ of its last two legs permutes S_3. Explicitly, for $C = C(r,r)$,

$$\bigotimes_{\pi \in S_3} ((C^{\pi})^{\otimes 2n/3}((C^{\sigma})^{\pi})^{\otimes n/3}) \cong Q_{r,r}^{\otimes n}.$$

This uses permutations of tensor factors within each global leg; it does not swap two global legs in only part of a tensor. Thus every separated balanced sector is exactly the prescribed product of the $Q_{r,b}$, up to a specified coordinate permutation.

Further finite refinements of this separator are proved in Appendix B. They are not used in the main extraction.

5 The counted finite construction

The parameters have separate roles. The profile index a determines the limiting exponent, and fixes the denominator scale K. The type scale N reduces the relative loss in counting sector words. The tensor power k amortizes interpolation, and the number j of exact reuses amortizes the catalyst. We choose them in this order.

Fix $a \ge 2$ and the integer types of Section 3.2. Index the concavity and tripling relations by disjoint sets $\mathcal{I}_c$ and $\mathcal{I}_t$, whose union is $\{1, \dots, I\}$. At an additional positive integer scale N, let the sector count of each relation be

$$M_i = \binom{Nw_i}{Nu_i} \quad \text{or} \quad M_i = \frac{(Nz_i)!}{(Nz_i/3)!^3}, \tag{21}$$

for a concavity or tripling relation, respectively. Define

$$M = \prod_{i=1}^{I} M_i, \qquad S = \prod_{i=1}^{I} (2M_i - 1)^6, \qquad H = M^6, \tag{22}$$

$$D = 6 \sum_{i=1}^{I} (M_i - 1)^2, \qquad u_0 = 2^{2NU}, \qquad v_0 = (2^{NK} M)^2. \tag{23}$$

The letters u_0, v_0 denote matrix dimensions, not the split counts u_i, d_i.

Proposition 5.1 (Counted catalytic degeneration)**.** *The specified maps give a polynomial degeneration of degree at most D:*

$$S\mathcal{C}_N Q_{a,a}^{\otimes NK} \mathsf{M}_{u_0} \trianglerighteq H\mathcal{C}_N \mathsf{M}_{v_0}. \tag{24}$$

Proof. Apply Lemma 4.1 with counts (Nu_i, Nd_i) to each concavity relation, and Lemma 4.2 with counts $(Nz_i/3, Nz_i/3, Nz_i/3)$ to each tripling relation. Write this exact type map as T_i, the separator map as $\mathsf{S}_i(t)$, and the sector-specific coordinate permutations after separation as R_i. For each $\pi \in S_3$, the complete relation map is

$$\Phi_{i,\pi}(t) = \mathsf{R}_i^{\pi} \circ \mathsf{S}_i(t)^{\pi} \circ \left(\bigoplus_{\ell=0}^{2M_i-2} \mathsf{T}_i^{\pi} \right). \tag{25}$$

Appendix A specifies its target spaces and every coefficient, including the positive-degree terms. In particular, R_i is a fixed map on the whole target space. For every relation and every leg permutation, apply Proposition 4.3 along the leg that is shared after that permutation, with auxiliary dimension M_i. It consumes $2M_i - 1$ source copies and produces M_i independent copies of its sector tensor, each with a dot product of dimension M_i.

Tensoring these maps over the six permutations and all relations gives the direct-sum counts S and H. The convolution factors on the source are $\mathcal{C}_N Q_{a,a}^{\otimes NK}$, and those on the output are $\mathcal{C}_N Q_{1,2}^{\otimes NK}$, by (16). The neighboring-length sources contribute NU dot products of dimension two in each of the six permutations. Their product is $\mathsf{M}_{2^{2NU}}$. The separator outputs contribute M_{M^2}, while $Q_{1,2}^{\otimes NK} \cong \mathsf{M}_{2^{2NK}}$. Equation (1) gives the stated dimensions.

The full map is $\bigotimes_{i,\pi} \Phi_{i,\pi}(t)$, followed by the displayed regroupings of factors. Use one common parameter in all separators. Lemma 2.1 bounds its degree by D. The initial type restrictions and final coordinate permutations are independent of t, so there are no further degree terms. □

The copy ratio satisfies $1 < S/H < 2^{6I}$. This inequality bounds a cost; it does not cancel either direct-sum multiplicity.

5.1 Exactification and repeated use of one catalyst

Define the explicit rank bound

$$\kappa_N = \prod_{h \le b} (h + b - 1)^{6N c_{h,b}} \ge \mathrm{R}(\mathcal{C}_N). \tag{26}$$

It comes from tensoring the decompositions (3), not from a rank-minimization oracle.

Proposition 5.2 (An exact relative multiplication scheme)**.** *For any positive integers k, j, put $E = kD + 1$, $A = u_0^{kj}$, $B = v_0^{kj}$, and*

$$r = \kappa_N^k E^j S^{kj} (2a - 1)^{6NKkj}, \qquad m = H^{kj}. \tag{27}$$

There is a specified exact restriction $r\mathsf{M}_A \trianglerighteq m\mathsf{M}_B$.

Proof. Apply Lemma 2.2 to the k-th power of (24). Replace each source copy of $Q_{a,a}$ by its explicit rank-$(2a-1)^6$ decomposition. Writing

$$\alpha = ES^k(2a-1)^{6NKk}, \qquad \beta = H^k,$$

we obtain the exact restriction

$$\alpha \mathcal{C}_N^{\otimes k} \mathsf{M}_{u_0^k} \succeq \beta \mathcal{C}_N^{\otimes k} \mathsf{M}_{v_0^k}.$$

This restriction may be applied j times while retaining one catalyst:

$$\alpha^j \mathcal{C}_N^{\otimes k} \mathsf{M}_{u_0^{kj}} \succeq \beta^j \mathcal{C}_N^{\otimes k} \mathsf{M}_{v_0^{kj}}.$$

For the induction, tensor the previous restriction with $\alpha \mathsf{M}_{u_0^k}$, then apply the one-step restriction to the retained catalyst in each output summand. Tensor products of matrix multiplication tensors combine by (1). The catalyst exponent stays k, not kj.

Supply its source using κ_N^k rank-one tensors and restrict the output catalyst to one nonzero coefficient. A coefficient equal to one is available by choosing a monomial in each convolution factor. This gives (27). □

Exactification precedes reuse. Iterating the degeneration itself would require a separate bound on its accumulated approximation degree.

5.2 Batched recursion with padding

Lemma 5.3 (Relative schemes give recursive algorithms)**.** *Suppose an explicit restriction $r\mathsf{M}_A \succeq m\mathsf{M}_B$ is given, with positive integers $B > A$. Put*

$$b_0 = \lceil r/m \rceil, \qquad p_0 = \frac{\log b_0}{\log(B/A)}.$$

Then $p_0 \geq 2$. The restriction yields a fixed arithmetic algorithm for matrix multiplication with cost $O(n^{p_0})$ if $p_0 > 2$, and $O(n^2 \log n)$ if $p_0 = 2$. Its scalar products are products of a linear form in the left input entries and a linear form in the right input entries.

Proof. The first-leg flattenings have ranks rA^2 and mB^2. Rank cannot increase under restriction, so $r/m \geq (B/A)^2$, which proves $p_0 \geq 2$. Compute a batch of m products of $n \times n$ matrices. Pad each input to side $B\lceil n/B \rceil$ and divide it into B-by-B blocks. The restriction reduces the batch to r matrix products of side $n' = A\lceil n/B \rceil$. The scalar identity remains valid on matrix blocks: each term keeps a left-input factor before a right-input factor, and its coefficients are central scalars. Group the resulting products into b_0 batches of size m, padding the last batch with zeros, and recurse. All preprocessing and output combinations at a node take $O(n^2)$ arithmetic operations, with constants depending only on the fixed scheme. Thus the batch cost satisfies

$$T(n) \leq b_0 T(A\lceil n/B \rceil) + Cn^2$$

above the fixed leaf threshold, for a constant C.

Set $\rho = A/B < 1$ and $c = A/(1-\rho) = AB/(B-A)$. Then $n' \leq \rho n + A$, or $n' - c \leq \rho(n-c)$. Choose the integer leaf threshold $n_* = \lceil c \rceil + 1$, and use classical multiplication at leaves. At level ℓ, each dimension is at most $c + \rho^\ell(n-c)$, so there are $O(\log_{1/\rho} n)$ levels. Summing the work over levels gives the two stated bounds. More explicitly, the part proportional to n^2 is bounded by a constant times $n^2 \sum_\ell (b_0\rho^2)^\ell$; the remaining constant-size contribution has the same or smaller order. The number of leaves is $O(n^{p_0})$.

Set all but one input pair of the initial batch to zero to obtain one matrix product. The fixed batch size changes only the constant. At each recursive call the input transformations are separately linear in the original left and right entries. This property persists to the classical products at the leaves, proving the last statement. □

In the present construction $r/m \geq 1$. Since $\lceil r/m \rceil \leq 2r/m$, Proposition 5.2 gives the upper expression

$$p_0 \leq \frac{6NK \log(2a-1) + \log(S/H) + \log(kD+1)/k + \log \kappa_N / j + \log 2/(kj)}{\log(v_0/u_0)} \tag{28}$$

whenever $v_0 > u_0$. Every copy, interpolation, catalyst, and batching charge is present. The source M_A remains the recursive operation; no fast decomposition of it has been assumed.

5.3 Finite parameter selection

Lemma 5.4 (A quantitative type bound)**.** *Let $n_1 + \cdots + n_s = n$, with all $n_i > 0$, and put $q_i = n_i/n$. Then*

$$\frac{\exp(nH(q))}{(n+1)^{s-1}} \leq \frac{n!}{\prod_i n_i!} \leq \exp(nH(q)).$$

Proof. The multinomial probability of the type (n_i) is its coefficient times $\prod q_i^{n_i} = \exp(-nH(q))$, and is at most one. It is also a mode. To see this, the factor $n_i^h/h!$ is maximized at $h = n_i$ or $h = n_i - 1$; compare consecutive terms. The joint type probability, up to a common factor, is the product of these factors. There are at most $(n+1)^{s-1}$ types, so a mode has probability at least their reciprocal. This proves both bounds. □

Put $\Delta_N = \log(v_0/u_0)$. Equations (18) and (23) imply

$$\begin{aligned} 2NK \log F(a,a) - 2\mathcal{E}_N \leq \Delta_N \leq 2NK \log F(a,a), \\ \mathcal{E}_N = \sum_{i \in \mathcal{I}_c} \log(Nw_i + 1) + 2 \sum_{i \in \mathcal{I}_t} \log(Nz_i + 1). \end{aligned} \tag{29}$$

For fixed a, the error is $O_a(\log N)$, while the main term is linear in N. As $\log(S/H) < 6I \log 2$, taking N large, then k large, and finally j large in (28) gives the limit

$$\tau_a = \frac{3 \log(2a-1)}{\log F(a,a)} \longrightarrow \frac{9}{4}. \tag{30}$$

For each finite a, $\tau_a > 9/4$ by Theorem 3.2.

The limiting argument can be replaced by the following precise selection procedure. Its comparisons involve only rational numbers raised to integer powers; no fixed numerical precision is assumed.

Theorem 5.5 (Effective finite extraction)**.** *There is a terminating procedure that, given a positive rational ε, specifies a finite arithmetic algorithm for multiplying arbitrary complex $n \times n$ matrices in $O_\varepsilon(n^{9/4+\varepsilon})$ operations. All coefficients lie in an explicitly specified cyclotomic field. The procedure uses only finite integer and rational computations, coordinate permutations, tensor products, and the maps proved above.*

Proof. Write $\theta = 9/4 + \varepsilon = p/q$ in positive integers. Perform the following searches in increasing integer order, stopping at the first successful choice in each step.

1. Choose $a \geq 2$ such that $F(a,a)^p > (2a-1)^{3q}$. Compute the rational weights, their least denominator K, the integer types, and the catalyst counts.

2. Choose $N \geq 1$. Compute $M_i, S, H, D, u_0, v_0, \kappa_N$ and put $R_0 = (S/H)(2a-1)^{6NK}$. Require
$$(v_0/u_0)^p > R_0^q. \tag{31}$$

3. With these data fixed, choose $k \geq 1$, put $E = kD+1$, and require
$$(v_0/u_0)^{pk} > (ER_0^k)^q. \tag{32}$$

4. Choose $j \geq 1$ such that
$$(v_0/u_0)^{pkj} > \big(2\kappa_N^k (ER_0^k)^j\big)^q. \tag{33}$$
Use Proposition 5.2, followed by Lemma 5.3.

Clear positive denominators to implement each test as an integer comparison. The first search terminates by (30). For its chosen a, (29) and the bounded copy-ratio charge imply (31) for all sufficiently large N. In particular $v_0 > u_0$, since $R_0 > 1$. In the third search the strict exponential advantage in (31) dominates $E^q = (kD+1)^q$. Once k is fixed, the strict exponential advantage in (32) dominates $(2\kappa_N^k)^q$ as j increases. Thus every search terminates; their order keeps all previously chosen data fixed.

The ratio of the actual counts in (27) is $r/m = \kappa_N^k (ER_0^k)^j$. Hence (33) implies $\lceil r/m \rceil^q < (B/A)^p$. The recursive exponent is strictly less than θ. Any quadratic or quadratic-logarithmic term in Lemma 5.3 is also $O(n^\theta)$, because $\theta > 2$.

To specify the maps, enumerate tensor positions and sector words in lexicographic order. Use the displayed determinant bases, interval sectors, Fourier substitutions, evaluation decompositions, and root averages. Separate sectors before applying their individual coordinate permutations. Compose the maps in the order in the proofs of Propositions 5.1 and 5.2. All roots of unity have explicitly known finite orders. One may take
$$L_* = \operatorname{lcm}\big(E,\, 2a-1,\, \{2M_i - 1 : 1 \leq i \leq I\},\, \{h+b-1 : c_{h,b} > 0\}\big).$$
Every coefficient lies in $\mathbb{Q}(\zeta_{L_*})$; catalyst reuse and block recursion introduce no further roots of unity. Thus the construction specifies finite coefficient arrays as well as their composition, even though expanding them is unnecessary for the proof. □

5.4 Rational coefficients and an explicit square seed

The field issue does not require an existence theorem for algebraic solutions of rank equations.

Lemma 5.6 (Restriction of coefficients to the rationals)**.** *Let T have rational coefficients, and suppose an explicit rank-R decomposition of T is given over a number field $\mathbb{K}$ of degree d. Then an explicit rational decomposition has length at most d^2R.*

Proof. Choose a rational basis $e_1 = 1, e_2, \dots, e_d$ of $\mathbb{K}$, and let $\phi : \mathbb{K} \to \mathbb{Q}$ extract the coefficient of e_1. For each summand write its first two linear forms as $\sum_s e_s \alpha_s$ and $\sum_t e_t \beta_t$, where the α_s, β_t have rational coefficients, and call its third form γ. Apply ϕ coefficientwise to the tensor identity. This summand becomes $\sum_{s,t} \alpha_s \beta_t \phi(e_s e_t \gamma)$, a sum of at most d^2 rational rank-one tensors. The target is unchanged because its coefficients are rational and $\phi(1) = 1$. All field operations are effective; for the field above, use the power basis modulo the cyclotomic polynomial. □

Corollary 5.7 (Finite rational seeds)**.** *For every positive rational ε, a terminating procedure produces an integer $d_0 \geq 2$ and a rational rank decomposition of M_{d_0} of length $R_{\text{seed}} < d_0^{9/4+\varepsilon}$. Consequently a fixed rational bilinear recursion attains an exponent strictly below $9/4 + \varepsilon$.*

Proof. Apply Theorem 5.5 with $\varepsilon/2$. For each integer dimension n, unroll its batched recursion to classical leaves, and set the unused initial input pairs to zero. Lemma 5.3 gives a bilinear decomposition of M_n. Its number of leaf products is at most its arithmetic cost, hence is $O_\varepsilon(n^{9/4+\varepsilon/2})$. Its coefficient field is fixed independently of n. Apply Lemma 5.6; this multiplies the length by at most a fixed constant.

Enumerate $n = 2, 3, \ldots$, generate the finite recursion and its rational decomposition, and stop when its length R_n satisfies $R_n^q < n^p$, where $p/q = 9/4 + \varepsilon$. The strict gap in exponents proves termination, regardless of the fixed constants. This is a construction of identities by proved operations, not a search for solutions to unspecified rank equations. Ordinary block recursion on the resulting seed has exponent $\log_n R_n < 9/4 + \varepsilon$; its fixed linear maps cost quadratically per node. □

Every rational seed identity also holds over every field of characteristic zero. The preprocessing time of this procedure is finite for each fixed ε. No bound uniform in ε, and no useful bit-complexity estimate, is asserted.

For $0 < \varepsilon \leq 3/4$, this prescribed seed generator returns $d_0 > n_* > A$, where A, n_* are chosen with allowance $\varepsilon/2$. At dimensions $n \leq n_*$, it produces the classical decomposition with n^3 nonzero terms, which fails the strict stopping test. Termwise coefficient descent preserves these already rational terms. This size constraint concerns the stated generator, not rational seeds in general.

6 Finite size and limitations of the prescribed construction

The preceding proof is effective, but its parameters grow rapidly. This section gives an exact denominator formula for every profile index and a finite example. Appendix C proves a lower bound for one prescribed small-index schedule. These size bounds concern the specified construction, not general matrix multiplication algorithms.

6.1 The minimal denominator scale

Theorem 6.1 (Exact normalization)**.** *The least integer scale in Section 3.2 is*

$$K_a = \operatorname{lcm}\left(\frac{3(a-1)(3a-1)}{\gcd(3a-1,4)},\quad \left\{\frac{3t(3t+4)}{\gcd(t,2)^2} : 1 \leq t \leq a-2\right\}\right). \tag{34}$$

In particular,

$$3\operatorname{lcm}(1,\ldots,a-1) \mid 2K_a, \qquad K_a \mid 3\operatorname{lcm}(1,\ldots,3a-1), \tag{35}$$

and

$$K_a \geq \frac{3}{2}\operatorname{lcm}(1,\ldots,a-1) \geq \frac{3\,2^{a-2}}{a}. \tag{36}$$

Proof. In the final row put $t = a-1$, $D_r = 3t+2$, and $b = a+h$. All split weights have common denominator $3tD_r$. Their integer numerators are D_r, from $\mu/3$, and $(D_r + 2h \pm 2)(D_r - h)$ for $0 \leq h \leq D_r - 1$. Modulo D_r, the latter products are $-2h(h \pm 1)$, all divisible by four. The case $h = 1$ with the plus sign is congruent to -4. Thus the joint gcd of the denominator and all numerators is exactly $\gcd(D_r, 4)$. This gives the first entry in (34).

In a lower row put $t = r - 1$ and $V = 3t + 4$. Use common denominator $3tV$. At $b = r$, the two split numerators are V^2 and $3tV$; the tripling split numerator is $2(3t+2)$. For $b = r + h > r$, the

split numerators are $2(3t+2+2h\pm 2)(3t+2-h)$. An odd prime dividing both V^2 and $2(3t+2)$ would divide the difference between V and $3t+2$, namely two, which is impossible. If t is odd, V^2 is odd, so the joint gcd is one. If t is even, all numerators and the denominator are divisible by four. When $t \equiv 2 \pmod 4$, V^2 has exactly two factors of two; when $t \equiv 0 \pmod 4$, $2(3t+2)$ has exactly two. Thus the joint gcd is exactly four. This proves the remaining entries.

Each row denominator is either $3\,\mathrm{lcm}(t, 3t+c)$ or half that number, with $c = 2$ for the final row and $c = 4$ for lower rows. The integers involved lie between one and $3a-1$; twice each row denominator is divisible by $3t$. Taking least common multiples proves (35).

For completeness, every binomial coefficient $\binom{n}{h}$ divides $\mathrm{lcm}(1,\dots,n)$. In its factorial valuation at a prime p, each power $p^j \le n$ contributes zero or one, so its valuation is at most $\lfloor \log_p n \rfloor$. The largest binomial coefficient in row n is at least $2^n/(n+1)$. Apply this with $n = a-1$ and use (35) to obtain (36). □

The scale is minimal for these exact rational split weights. Approximate types, different weights, or different tensor maps need not obey this bound. Since $U \ge K_a$, already the unpowered source dimension $u_0 = 2^{2NU}$ grows at least doubly exponentially in a under this bound. The inequality $U \ge K_a$ also follows directly from (16): the net consumption of $[a,a]$ is K_a, and all its source occurrences belong to concavity relations.

6.2 Small indices and a finite example

Table 1 gives exact counts from the rational formulas. The values of τ_a are displayed approximately.

a	K_a	U	T	$U+T-K_a$	τ_a
2	15	230	15	230	2.737468
3	84	3316	162	3394	2.641126
4	3465	252362	8877	257774	2.592310

Table 1: The normalization, source type counts, catalyst factor count, and limiting exponent for the first three indices. The exponent column is a limit for fixed a, not the cost of these unamortized types.

For $a = 2$, the concavity types, indexed by $b = 2,\dots,6$, are

b	2	3	4	5	6
w_b	50	56	54	44	26
u_b	35	36	33	26	15
d_b	15	20	21	18	11

The tripling count is $z = 15$. In this subsection write $Q_b = Q_{2,b}$. The catalyst is

$$\mathcal{C}_N = Q_2^{35N} Q_3^{56N} Q_4^{54N} Q_5^{44N} Q_6^{26N} Q_7^{15N},$$

where powers denote tensor powers. Hence

$$\kappa_N = 3^{210N} 4^{336N} 5^{324N} 6^{264N} 7^{156N} 8^{90N}, \qquad u_0 = 2^{460N}, \quad v_0 = 2^{30N} M^2. \tag{37}$$

The entropy identity becomes

$$\frac{3^{15}}{2^{215}} \prod_{b=2}^{6} \frac{w_b^{w_b}}{u_b^{u_b} d_b^{d_b}} = (10/3)^{15}.$$

The archived calculation for $(N, k, j) = (300, 125, 3750)$ bounds (28), using $\log(S/H) < 36 \log 2$, in the interval $(2.78918, 2.78919)$. Its target dimension $B = v_0^{kj}$ has 21 665 218 118 decimal digits; the recursive input dimension $A = u_0^{kj}$ has 19 472 877 845 digits. These are finite construction parameters, not measured running times or storage requirements of an implemented algorithm. The example illustrates exact finite cost accounting; it is not competitive with existing theoretical bounds or practical methods.

The saved certificates use rational logarithm enclosures. For clarity, the entire enclosure method is short. Write a positive integer as $2^e y$, $1 \le y \le 2$, bracket y by adjacent rationals with denominator 2^{80}, and set $z = (y-1)/(y+1)$. Then

$$2 \sum_{h=0}^{39} \frac{z^{2h+1}}{2h+1} \le \log y \le 2 \sum_{h=0}^{39} \frac{z^{2h+1}}{2h+1} + \frac{2z^{81}}{81(1-z^2)}. \tag{38}$$

The bound follows by summing the positive tail of the power series and using $2h+1 \ge 81$ in that tail. Apply the same bound to $\log 2$, then combine rational intervals. For a positive rational, enclose the logarithms of its numerator and denominator separately and subtract the intervals. Decimal digit counts follow by enclosing $\log_{10} A$ or $\log_{10} B$ strictly between consecutive integers. These saved computations corroborate the displayed example; no numerical experiment is required by Theorem 5.5.

Appendix C proves an actual-count lower bound for this prescribed schedule, independent of how its three parameters are tuned.

7 Consequences, scope, and further questions

The construction gives an effective answer to a particular extraction problem. The convolution relations can be combined with probabilities and integer multiplicities that are known in advance. Their composition produces an exact relative multiplication identity, and a concrete recursion converts that identity to ordinary matrix multiplication. Neither a detecting tensor character nor an unknown optimal rank decomposition is needed. Theorem 5.5 and Corollary 5.7 make this statement precise for each fixed positive exponent allowance.

The least-profile theorem has a different use. It identifies the exact limit of arguments using only normalization, symmetry, concavity, and shifted tripling. A stronger diagonal exponent would require additional constraints, rather than more instances of the same scalar inequalities. The diagonal bound already follows from tripling at the row index, which is the subfamily used in the tensor construction. Only the off-diagonal minimality proof needs all the shifted tripling inequalities. The rational certificate is useful beyond its numerical bound because it specifies precisely which tensor factors must be supplied, produced, and retained. This makes hidden cancellation assumptions testable at the level of finite maps.

The separator refinements also have an independent scope. They apply to any family of matched sectors sharing one leg. The smaller copy count and the two-power projection improve particular finite reductions, without requiring any convolution structure. Their use in another tensor argument would still require separate accounting for the auxiliary dot products and for conversion of simultaneous outputs to an algorithm.

There are three limitations to the exponent conclusion. First, every finite profile index in this construction has $\tau_a > 9/4$; the proof does not supply one fixed recursion with bound $O(n^{9/4})$. Second, the algorithm and coefficient set are fixed after the allowance is chosen. A uniform version with a changing allowance requires additional conventions for generating and loading its rational constants, as well as an analysis of setup time. We do not give a quantitative convergence rate for

such a version. Third, its exact arithmetic identities do not establish numerical stability or useful finite-size performance. The denominator growth and the small-index cost obstruction explain the large parameter sizes of the prescribed construction. They do not give a stability analysis.

Several changes could alter this conclusion about cost. One could replace the exact tangent probabilities by types whose error is controlled with smaller denominators; use a different family of convolution relations; find a cheaper way to supply the common tensor factors; or exploit structure among the simultaneous recursive products. None of the lower bounds in Section 6 or Appendix C excludes these alternatives. Establishing such a change would require a new finite identity or a new cost analysis, not an interpretation of an asymptotic tensor inequality as a finite cancellation rule.

Supporting calculations and source files

The proof is contained in the formulas and arguments in this paper. The accompanying supplement, `finite-extraction-supplement-v1.zip` (version 1, 7 October 2026), includes exact-arithmetic scripts for the rational weights and resource counts, component-map checks, and saved rational interval certificates. The general type data are generated by `experiments/general_profile_extraction.py`; the small-index certificates are recorded in `results/small-profile-extraction.json` and their method in `experiments/small_profile_extraction.py`. The separator calculations are in `experiments/finite_constructions.py`, and the scalar-profile calculations in `experiments/profile_bounds.py`. These files are supplementary checks, not assumptions of the main theorem. The complete composed coefficient arrays are specified by Theorem 5.5; they are not stored as expanded arrays.

The numerical example and interval bounds in Section 6 and Appendix C are taken from the archived calculations. They were not obtained from timing a realization of the large construction. No numerical benchmark is used to infer an arithmetic exponent.

A The full composed relation maps

This appendix writes the composition in (25) on its full variable spaces. In particular, a sector-specific reindexing is a fixed linear map even on the positive-degree terms, whose sector labels need not agree.

Fix the scale N of Section 5. For relation i, let its source tensor and its canonical sector tensor be

$$\mathcal{U}_i = \begin{cases} (C(r,b)B_X(2))^{\otimes N w_i}, & i \in \mathcal{I}_c, \\ C(r,4r-1)^{\otimes N z_i}, & i \in \mathcal{I}_t, \end{cases}$$

and

$$\mathcal{V}_i = \begin{cases} C(r,b+1)^{\otimes N u_i} C(r,b-1)^{\otimes N d_i}, & i \in \mathcal{I}_c, \\ C(r,r)^{\otimes 2N z_i/3} (C(r,r)^{\sigma})^{\otimes N z_i/3}, & i \in \mathcal{I}_t, \end{cases}$$

where σ exchanges the second and third legs. In the tripling case, the exchanged factors remain in this expression until the six permutations are combined.

Let T_i be the exact type map of Lemma 4.1 or Lemma 4.2. Enumerate its retained words by $h = 0, \dots, M_i - 1$, and write

$$\mathsf{T}_i(\mathcal{U}_i) = \sum_{h=0}^{M_i-1} \sum_{\alpha,\beta,\gamma} c_{i,h}(\alpha,\beta,\gamma) x_\alpha y_{h,\beta} z_{h,\gamma}.$$

The first variable space X_i is shared. The other two spaces are $\bigoplus_h Y_{i,h}$ and $\bigoplus_h Z_{i,h}$, and the indices β, γ range over the coordinates of sector h.

Set $L_i = 2M_i - 1$, and let $\mathsf{S}_i(t)$ be the substitutions of Proposition 4.3 with $K_0 = M_i, c = 1$. After the type maps in the L_i source copies, these give exactly

$$\mathsf{S}_i(t)\left(\bigoplus_{\ell=0}^{L_i-1} \mathsf{T}_i\right)(L_i\mathcal{U}_i) = \sum_{\substack{0\le h,g,u,v<M_i \\ u-v+g-h=0}} t^{(g-h)^2} \sum_{\alpha,\beta,\gamma} c_{i,h}(\alpha,\beta,\gamma) X_{\alpha,g} Y_{h,\beta,u} Z_{h,\gamma,v}. \tag{39}$$

The Fourier sum imposes the displayed equality, and the combined weight is $(g-h)^2$. This formula includes every positive-degree term.

For example, for a neighboring-length relation put $n = Nw_i$, and let $\omega(h) \in \{+,-\}^n$ be its retained word. In the quotient and kernel bases, before reordering positions,

$$c_{i,h}(\boldsymbol{\alpha},\boldsymbol{\beta},\boldsymbol{\gamma}) = \prod_{\nu=1}^{n} \mathbf{1}_{\{\gamma_\nu = \alpha_\nu + \beta_\nu\}}.$$

Here $0 \le \alpha_\nu < r$. At a $+$ position, $0 \le \beta_\nu \le b$ and $0 \le \gamma_\nu \le r+b-1$; at a $-$ position, $0 \le \beta_\nu \le b-2$ and $0 \le \gamma_\nu \le r+b-3$. Thus the coefficient of $X_{\boldsymbol{\alpha},g} Y_{h,\boldsymbol{\beta},u} Z_{h,\boldsymbol{\gamma},v}$ in (39) is

$$\mathbf{1}_{\{u-v+g-h=0\}}\, t^{(g-h)^2} \prod_{\nu=1}^{n} \mathbf{1}_{\{\gamma_\nu=\alpha_\nu+\beta_\nu\}}.$$

For a tripling relation the coefficient is likewise a product of $\mathbf{1}_{\{\gamma_\nu=\alpha_\nu+\beta_\nu\}}$, with the original interval coordinates in Lemma 4.2; the middle-position reversals are part of the subsequent reindexing.

The separator target spaces are

$$\widetilde{X}_i = \bigoplus_g X_i^{(g)}, \qquad \widetilde{Y}_i = \bigoplus_h (Y_{i,h} \otimes \mathbb{C}^{M_i}), \qquad \widetilde{Z}_i = \bigoplus_h (Z_{i,h} \otimes \mathbb{C}^{M_i}).$$

For each h, let $(\mathsf{R}^X_{i,h}, \mathsf{R}^Y_{i,h}, \mathsf{R}^Z_{i,h})$ be the coordinate permutations identifying its sector tensor with $\mathcal{V}_i$. They group tensor positions by type; in the tripling case they also shift each second- and third-leg interval to start at zero and reverse the first index at middle positions. Define the maps on the full target spaces by

$$\mathsf{R}_i = \left(\bigoplus_h \mathsf{R}^X_{i,h}, \quad \bigoplus_h (\mathsf{R}^Y_{i,h} \otimes I_{M_i}), \quad \bigoplus_h (\mathsf{R}^Z_{i,h} \otimes I_{M_i})\right).$$

The first component acts on block h of $\widetilde{X}_i$, whether or not a monomial's other sector labels equal h. Hence all three maps are defined on every term of (39), and they are independent of t.

The map $\Phi_{i,\pi}(t)$ of (25) acts on $L_i\mathcal{U}_i^\pi$. Its separator acts along the leg occupied by X_i after permutation by π. The constant term in (39) has $g = h$ and $u = v$, so

$$\Phi_{i,\pi}(t)(L_i\mathcal{U}_i^\pi) = \bigoplus_{h=0}^{M_i-1} (\mathcal{V}_i B_X(M_i))^\pi + \sum_{d=1}^{(M_i-1)^2} t^d G_{i,\pi,d}.$$

In particular, the constant term has M_i full direct-sum copies.

Tensor $\Phi_{i,\pi}(t)$ over all i, π, using one parameter. The source and constant-term multiplicities are $\prod_i L_i^6 = S$ and $\prod_i M_i^6 = H$. Lemma 2.1 bounds the degree by $6\sum_i (M_i-1)^2 = D$. Regroup the source factors and the constant-term factors by coordinate permutations independent of t. For tripling, the mixed factors $\mathcal{V}_i^\pi$ combine over all six permutations to $Q_{r,r}^{\otimes N z_i}$, by the orientation-balance identity in Section 4. The ledger (16) and the dot-product identities (1) then give exactly (24).

B Sparse exactification and a two-power refinement

These refinements explain which costs of a separator can be reduced. The general extraction in Section 5 uses the simpler degree bound of Lemma 2.2. If $G(t) = \sum_{s\in E} t^s G_s$, with finite nonnegative integer support E containing zero, define

$$P(z) = \prod_{s\in E\setminus\{0\}} \frac{z-2^s}{1-2^s} = \sum_{j=0}^{|E|-1} \alpha_j z^j.$$

Then $\sum_j \alpha_j G(2^j) = G_0$. This gives an exact rational interpolation formula using $|E|$ evaluations. For a separator, each evaluation uses L source copies, for a total of $L|E|$. For one separator, applying this prescription to the guaranteed support set $\{cd^2 : 0 \le d \le d_*\}$ cannot improve on direct copying: the prescribed count satisfies

$$L(d_*+1) \ge MK_0. \tag{40}$$

If $d_* = M-1$, use $L \ge K_0$. Otherwise $d_*+1 = \lceil K_0/c \rceil$, and $L(d_*+1) \ge K_0(d_*+1) + K_0(M-1) \ge MK_0$. Direct copying uses one source for each pair (h,u), and restricts that source to sector h. Equation (40) concerns these particular copy counts, not all possible exactifications.

A smaller exact projection is possible for two powers of the separator polynomial G. If a prime q satisfies $q > d_*$, $q \equiv 3 \pmod 4$, and $q \nmid c$, then for a primitive q-th root η,

$$\frac{1}{q}\sum_{j=0}^{q-1} G(\eta^j)^{\otimes 2} = G_0^{\otimes 2}. \tag{41}$$

Indeed a surviving exponent would require $d_1^2 + d_2^2 \equiv 0 \pmod q$. Since -1 is not a square modulo q, both d_i vanish modulo q, hence are zero. For $M = K_0 = 8, c = 1$, one can use $L = 15, q = 11$, giving $11 \cdot 15^2 = 2475$ source copies for the squared tensor, compared with $8^4 = 4096$ by direct copying. No unbounded prime search is needed for the main result: the general degree projection is always available.

C An actual-count lower bound for the small-index schedule

The following result concerns exactly the interpolation count and catalyst decomposition in (27). Its hypothesis uses the actual branching count, rather than the sufficient upper bound (28). We use $\log_4 48$, the recursive exponent of a known rational rank-48 algorithm for 4×4 multiplication [7], as a fixed comparison value.

Proposition C.1 (A limitation of parameter tuning)**.** *For the $a = 2$ construction, use $E = kD+1$, the catalyst bound (37), and the full-batch recursion of Lemma 5.3. Assume $B > A$. If*

$$\frac{\log\lceil r/m\rceil}{\log(B/A)} < \rho_{48} := \frac{\log 48}{\log 4},$$

then its recursive input dimension A has more than 11.37 billion decimal digits. One of its separator sector counts also exceeds 10^{1206}.

Proof. Use the small-index counts above and define

$$\begin{aligned}
\delta &= 30\rho_{48}\log(10/3) - 90\log 3,\\
\gamma &= 6(35\log 3 + 56\log 4 + 54\log 5 + 44\log 6 + 26\log 7 + 15\log 8),\\
\mu &= \log\binom{56}{36}.
\end{aligned}$$

Thus $\log \kappa_N = N\gamma$. The hypothesis and $\lceil r/m \rceil \geq r/m$ imply the necessary condition

$$\frac{\log E}{k} + \frac{N\gamma}{j} < \rho_{48}\Delta_N - 90N \log 3. \tag{42}$$

We have dropped the positive copy-ratio charge, making the condition weaker.

The one-sided Stirling estimate [6, 11] $\log(n!) = (n+\frac{1}{2})\log n - n + \frac{1}{2}\log(2\pi) + r_n$, $0 < r_n < 1/(12n)$, gives

$$\Delta_N \leq 30N \log(10/3) - 7 \log N - 2C_0 + \frac{2\eta}{N}, \tag{43}$$

where

$$C_0 = \frac{1}{2}\sum_{b=2}^{6} \log(6u_b d_b/w_b) + \log 6 + \frac{3}{2}\log 5 - \frac{1}{2}\log 15,$$
$$\eta = \sum_{b=2}^{6} \frac{1}{12w_b} + \frac{1}{180}.$$

For each binomial coefficient, subtract the three Stirling formulas and keep only the positive remainder $1/(12Nw_b)$. Its correction to entropy is at most $-\frac{1}{2}\log N - \frac{1}{2}\log(2\pi u_b d_b/w_b) + 1/(12Nw_b)$. The three-sector coefficient contributes $-\log N - \log(2\pi) + \frac{1}{2}\log 15 - \frac{3}{2}\log 5 + 1/(180N)$. Replace 2π by the smaller number six and use the exact entropy identity to obtain (43).

The right side of (42) is therefore at most

$$f(N) = \delta N - \rho_{48}(7 \log N + 2C_0 - 2\eta/N).$$

Its second derivative is $\rho_{48}(7/N^2 + 4\eta/N^3) > 0$. The archived rational enclosures from (38) give $f(1) < -71.4589$ and $f(80) < -0.224635$. A convex function is at most its endpoint chord on $[1, 80]$. The necessary condition has a positive left side, so $N \geq 81$. The penalty $7 \log N + 2C_0 - 2\eta/N$ is increasing. Put $\beta = \rho_{48}(7 \log 81 + 2C_0 - 2\eta/81)$. Then

$$\frac{\log E}{k} + \frac{N\gamma}{j} < \delta N - \beta. \tag{44}$$

Selecting 36 positions within each of N disjoint blocks of 56 proves $M_3 = \binom{56N}{36N} \geq \binom{56}{36}^N$. Because $M_3 \geq 2$, $E \geq 6(M_3 - 1)^2 + 1 \geq M_3^2$, hence $\log E \geq 2N\mu$. Arithmetic–geometric mean in (44) yields

$$Nkj > \frac{8\mu\gamma N^3}{(\delta N - \beta)^2} \geq \frac{54\mu\gamma\beta}{\delta^3}.$$

Here $\delta N > \beta$, and the last inequality follows from $4x^3 - 27(x-1)^2 = (x-3)^2(4x-3) \geq 0$, $x = \delta N/\beta > 1$. The same saved rational enclosures certify

$$\frac{54\mu\gamma\beta}{\delta^3} > 82\,111\,374.98, \qquad 460\frac{54\mu\gamma\beta}{\delta^3}\frac{\log 2}{\log 10} > 11\,370\,273\,954.02, \qquad 81\frac{\mu}{\log 10} > 1206.$$

Since $A = 2^{460Nkj}$, these inequalities prove both assertions. □

The conclusion does not apply after pruning zero branches, replacing the catalyst decomposition, sharing bilinear work, or changing exactification. Those operations change the counted schedule. Factorized Fourier circuits can reduce its linear work and storage, but by themselves do not reduce the independent source calls counted in (27).

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